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Abstract

Knowledge tracing is a core component of adaptive learning systems, yet much of the literature assumes binary correctness as the observation signal. In many educational settings, however, learner evidence is polytomous, including ordered score levels and partial credit, where both ordinal structure and interpretability matter for measurement and intervention.

We propose DEEP-MIRT, a modular framework that extends knowledge tracing to polytomous responses by bridging sequential neural modeling with psychometric Item Response Theory. The framework couples a swappable sequential backbone with an interchangeable response head, separating the question of how learner ability evolves over time from the question of how that ability maps to ordered response categories. Instantiated with a Generalized Partial Credit Model (GPCM) head, DEEP-MIRT produces psychometrically interpretable outputs at every interaction, including an evolving student ability estimate, item discrimination values, and ordered difficulty thresholds. The model delivers strong predictive performance while accurately recovers ground-truth item parameters across varying category resolutions. The framework advances knowledge tracing from a purely predictive instrument into a measurement-aware system that grounds instructional decision-making in interpretable psychometric theory.

Keywords: Knowledge tracing, Item response theory, Polytomous responses, Deep learning, Psychometrics

1 Introduction

Educational assessments often target graded performance rather than pass/fail outcomes. Rubric-based scoring, partial-credit items, and constructed-response tasks encode degrees of correctness that binary labels collapse into a single signal (Black

& Wiliam, 1998; Heritage, 2007). While automated scoring systems can deliver this ordinal structure at scale in intelligent tutoring contexts (Baral, Botelho, Erickson, Benachamardi, & Heffernan, 2021; Shute, 2011), most KT benchmarks, evaluation protocols, and modeling paradigms still assume binary correctness. This leaves ordinal outcomes comparatively underexplored in KT, even though they are central to authentic assessment practice.

Beyond the outcome type, interpretability remains a key concern. Most KT models learn high-dimensional states optimized for prediction, but these states are not, by construction, tied to interpretable quantities. Therefore, constructs that educators and assessment designers routinely work with, such as student ability level, item discriminatory power, or the difficulty steps between adjacent score categories, are not directly identifiable. Item Response Theory (IRT) provides exactly this structure through identifiable parameters including student ability, item discrimination, and ordered step difficulties, grounding score interpretation in established psychometric theory (Embretson & Reise, 2000; Messick, 1995; Muraki, 1997). Standard IRT models, however, treat ability as fixed and are not designed for sequential interaction data. This work therefore extends KT to polytomous responses by incorporating the interpretable parameter structure of IRT within a dynamic sequential framework.

Three lines of existing work address these issues in isolation, leaving a gap at their intersection. First, deep KT models, from Deep Knowledge Tracing (DKT; Piech et al., 2015) to memory-augmented and attention-based variants (Pandey & Karypis, 2019; Zhang, Shi, King, & Yeung, 2017), capture temporal learning dynamics but typically assume dichotomous correctness and learn latent states with limited psychometric interpretability. Second, polytomous Item Response Theory (IRT) models such as the Generalized Partial Credit Model (GPCM; Muraki, 1997) provide principled, interpretable measurement of ordered responses with identifiable parameters (student ability, item discrimination, ordered step thresholds), but they generally treat ability as time-invariant and are not designed for sequential interaction logs. Third, recent efforts that inject IRT structure into deep KT (Converse, Curi, Oliveira, & Templin, 2021; Yeung, 2019) improve interpretability but largely remain restricted to dichotomous outcomes. An early partial exception is Wang and Heffernan (2013), who extended Bayesian Knowledge Tracing to allow partial credit via continuous Bayesian nodes, but without deep sequential modeling or recovering standard polytomous IRT step structure. Yet these approaches has not yet reconciled sequential dynamics with ordinal response modeling while retraining psychometric identifiability.

We propose *DEEP-MIRT*, a modular framework that extends KT to Polytomous Knowledge Tracing (PKT) by coupling a swappable sequential backbone with an IRT-grounded response head. In this setting, PKT is the task of predicting the full distribution over ordered response categories from a learner's interaction history. We instantiate the framework with a GPCM response head, yielding DEEP-GPCM, which combines a Dynamic Key-Value Memory Network (DKVMN) backbone with a GPCM observation model. At each timestep, the GPCM head maps the learner state to a distribution over ordinal categories and exposes psychometrically interpretable components, including an evolving student ability estimate, as well as learned item

parameters such as per-item discrimination values and ordered step difficulties corresponding to score category transitions. While the experiments focus on ordinal PKT, the framework accommodates other polytomous formats through alternative response heads.

We make the following contributions and investigate corresponding research questions:

1. **A modular framework for polytomous knowledge tracing.** We propose DEEP-MIRT, which decouples a sequential backbone from an interchangeable psychometric response head. We instantiate the framework with a GPCM head (DEEP-GPCM), producing evolving ability trajectories, item discrimination, and ordered step difficulties—constructs that assessment designers already use in practice. We ask: does coupling a DKVMN backbone with a psychometrically constrained head improve ordinal prediction over format-unaware alternatives (**RQ1**), and do the inferred ability trajectories exhibit plausible learning dynamics (**RQ2**)?
2. **Psychometric parameter recovery from interaction data.** We evaluate whether DEEP-GPCM can recover ground-truth IRT parameters (θ, α, β) from synthetic data without direct supervision, and how recovery varies across parameter types and category resolutions (**RQ3**).
3. **Scalable item encoding and robustness analysis.** We propose Static Item Embedding (SIE) for large item banks and evaluate encoding strategies, monotonicity constraints, and class imbalance robustness to characterize the framework's operational boundaries.

2 Related Work

2.1 Knowledge Tracing: From Binary to Polytomous

Knowledge tracing has evolved through several modeling paradigms, but most of them assume a binary mastery state. Bayesian Knowledge Tracing (BKT) models mastery of a Knowledge Component (KC) as a two-state Hidden Markov Model (Corbett & Anderson, 1994). Deep Knowledge Tracing (DKT) reframed the task as sequence modeling with recurrent neural networks, learning high-dimensional latent states from interaction sequences (Piech et al., 2015). Dynamic Key-Value Memory Networks (DKVMN) introduced a structured memory that separates static concept representations (key memory) from evolving mastery states (value memory), offering a more transparent mapping between knowledge components and latent state than generic RNNs (Graves et al., 2016; Zhang et al., 2017). Attention-based models such as SAKT and AKT apply self-attention to weigh the relevance of prior interactions for next-step prediction (Ghosh, Heffernan, & Lan, 2020; Pandey & Karypis, 2019), and recent work has further explored Transformer-based architectures (Choi et al., 2020; Liu, Liu, Chen, Huang, & Luo, 2023). Throughout this progression, the response format has remained dichotomous: every model cited above predicts a binary correct/incorrect outcome.

Recent KT research has begun to acknowledge that correctness alone is insufficient. Lu et al. (2024) fuse process features (attempts, response time, historical accuracy) to

enrich difficulty representations, and Li et al. (2025) integrate problem complexity and state-stability mechanisms to reduce noisy updates. These efforts treat richer signals as auxiliary features within a binary prediction framework. They do not change the response format itself, but they highlight that educational interactions carry information beyond pass/fail: precisely the information that ordinal and polytomous responses encode directly. The only prior work to move beyond binary responses in a KT context is Wang and Heffernan (2013), who extended BKT to partial credit using continuous Bayesian nodes, but without deep sequential modeling or IRT-based parameterization.

2.2 Interpretability in Deep Knowledge Tracing

In most DKT models, learned states are computed as high dimensional representations than psychometrically meaningful constructs (National Research Council, 2001). Several efforts address this gap by coupling neural architectures with Item Response Theory (IRT) structure. Deep-IRT attaches an IRT-style readout to DKVMN, exposing student ability and item difficulty parameters for each prediction (Yeung, 2019). DKVMN-MRI similarly integrates IRT within an LSTM-based memory framework (Xu et al., 2024). Converse et al. (2021) propose a general approach for incorporating IRT parameters into KT training. These IRT-augmented models improve interpretability: a teacher or assessment specialist can inspect the estimated ability and difficulty values, rather than relying on opaque hidden states. In educational terms, "interpretability" here means that model outputs correspond to constructs that practitioners already use (student proficiency levels, item difficulty, and discrimination quality), supporting the kinds of validity arguments that responsible assessment requires (Kane, 2013; Messick, 1995).

A critical limitation, however, is that all existing IRT-KT integrations operate on dichotomous responses. They estimate a single ability and difficulty for binary outcomes and do not model the ordered step structure of graded scores or partial-credit items. Attention-based models provide a complementary form of interpretability through attention weights (which interactions were most relevant to a prediction), but they too lack psychometric parameterization and ordinal structure awareness.

2.3 Polytomous Item-Response Theory

Psychometrics provides a family of models for ordered responses grounded in a latent ability parameter θ and item-specific properties (Embretson & Reise, 2000; Hambleton & Swaminathan, 1985). Two traditions dominate for ordinal outcomes. Cumulative models such as the Graded Response Model (GRM; Samejima, 1969) define boundary parameters between adjacent categories, modeling the probability of scoring at or above category k as a logistic function of θ. Adjacent-category models, including the Partial Credit Model (PCM; Masters & Wright, 1997) and the Generalized Partial Credit Model (GPCM; Muraki, 1997), parameterize each step transition directly: the GPCM characterizes each item by a discrimination parameter α_j controlling how sharply the item differentiates ability levels, and ordered step difficulty thresholds $\boldsymbol{\beta}_j = (\beta_{j,0}, \ldots, \beta_{j,K-2})$ representing the ability level at which each consecutive category transition becomes equally likely. Adjacent-category models are the natural fit

for graded scoring, where performance progresses sequentially through ordered levels (Embretson & Reise, 2000; Wilson, 2005). For unordered categories, the Nominal Response Model (NRM; Bock, 1972) provides an analogous framework.

In the context of knowledge tracing, these static measurement paradigms must be reformulated to capture sequential learning dynamics. Formally, at time step t, a student interacts with an item indicated by a one-hot vector $q_t \in \{0,1\}^Q$ over Q unique items, yielding an ordinal response $r_t \in \{0, 1, \ldots, K-1\}$ across K ordered score levels. Defining an interaction as the tuple $x_t = (q_t, r_t)$ and the cumulative history as $\mathcal{X}_t = \{x_1, \ldots, x_t\}$, the objective of polytomous knowledge tracing is to predict the conditional distribution of the next response:

$$P(r_{t+1} = k \mid q_{t+1}, \mathcal{X}_t), \quad k \in \{0, \ldots, K-1\}. \tag{1}$$

This work therefore explores how the interpretable parameter structures of polytomous IRT might be adapted for sequential modeling. We treat it as a time-varying latent state conditioned on the interaction history $\mathcal{X}_t$, aiming to interpret dynamic tracing results with psychometric models.

3 Methodology

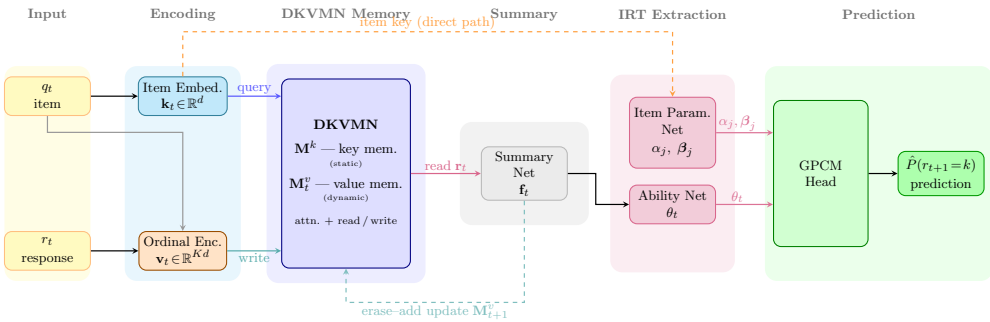

Fig. 1 DEEP-MIRT framework at timestep t. Item parameters $(\alpha_j, \boldsymbol{\beta}_j)$ bypass the student state via the item embedding (dashed, top), consistent with the IRT assumption that discrimination and difficulty are student-independent. The GPCM head maps $(\theta_t, \alpha_j, \boldsymbol{\beta}_j)$ to the predicted ordinal distribution $\hat{P}(r_{t+1} = k)$.

3.1 Framework Overview

DEEP-MIRT is a modular architecture with two separable concerns. The *backbone* handles the knowledge tracing problem: it encodes item–response interactions, maintains a dynamic memory of the student's evolving knowledge state, and produces a summary vector $\mathbf{f}_t$ representing current student state. The *response head* handles the measurement problem: it maps the backbone's student state $\mathbf{f}_t$ and item identity $\mathbf{k}_t$ to a probability distribution over response categories through a psychometric observation model. Any differentiable function $h(\mathbf{f}_t, \mathbf{k}_t) \to \Delta^{K-1}$ (a distribution over K

categories) satisfying this interface is a valid DEEP-MIRT instantiation. The backbone is agnostic to which measurement model governs the head.

This separation is conceptually and practically helpful for tracking learning progressions. The backbone addresses the learning dynamics question of how student ability evolves across a sequence of interactions, naturally handled by sequential neural architectures. The head addresses the measurement question of how current ability maps to the probability of each ordered score category, naturally handled by psychometric theory. Coupling them through a clean interface means the two concerns can be improved independently: a better backbone (e.g., Transformer instead of DKVMN) improves temporal modeling without changing the interpretable parameterization; a different head (e.g., GRM instead of GPCM) changes the measurement assumptions without retraining the backbone.

In this paper we instantiate the head with the GPCM over the other major polytomous families for three reasons: (1) it is an adjacent-category model, which is the natural match for graded scoring where a student progresses sequentially through performance levels; (2) it produces a discrimination parameter α_j and ordered step difficulty thresholds $\boldsymbol{\beta}_j$ that correspond directly to scoring design decisions and can be inspected by assessment practitioners; (3) its monotone threshold constraint admits a clean differentiable parameterization as described below. A cumulative model such as the GRM (Samejima, 1969) would be a natural alternative for tasks where each scoring threshold represents an independent decision (e.g., analytic rubrics with independent sub-scores); a NRM (Bock, 1972) would apply when response categories are unordered (e.g., multiple-choice representing different misconceptions). We treat these as directions for future instantiation.

The backbone processes inputs through four sequential stages as shown in Figure 1: ordinal interaction encoding, DKVMN memory update, summary compression, and IRT-based parameter extraction. An ordinal-aware training objective supervises all stages jointly. The following subsections formalize each component.

3.2 Ordinal-Aware Interaction Representation

Linear Decay Encoding

We encode an ordinal response r_t with a "triangular" kernel over ordered categories. Define a length-K ordinal code $\mathbf{s}_t \in \mathbb{R}^K$ by

$$\begin{aligned} s_t^{(k)} &= \max\left(0, 1 - \frac{|k - r_t|}{K-1}\right), \quad 0 \le k \le K-1, \\ \mathbf{s}_t &= [s_t^{(0)}; \ldots; s_t^{(K-1)}]. \end{aligned} \tag{2}$$

For example, with $K = 4$ score categories and a response of $r_t = 2$ ("correct approach with errors"), the ordinal code is $\mathbf{s}_t = [0.33,\ 0.67,\ 1.0,\ 0.67]$, which spreads activation to neighboring categories while peaking at the observed level. This encoding preserves the ordinal structure: a response of 2 is represented as more similar to responses 1 and 3 than to response 0, reflecting the educational intuition that partial credit is closer to full credit than to no credit.

We then bind the ordinal code to the item indicator via a Kronecker product (equivalently, a block one-hot expansion):

$$\mathbf{x}_t = q_t \otimes \mathbf{s}_t \in \mathbb{R}^{KQ}. \tag{3}$$

This reduces to the standard binary interaction encoding when $K = 2$ (consistent with Zhang et al. (2017)). While this encoding is effective for moderately sized item banks, the $\mathcal{O}(KQ)$ dimensionality becomes a practical constraint for large Q; we address this in the following subsection.

Scalable Value Encoding for Large Item Banks

Large-scale educational platforms maintain extensively large item banks with various grade levels. The triangular Kronecker encoding in (2) yields an interaction vector of dimension KQ, which grows linearly with item bank size. The subsequent value projection $\mathbf{v}_t = B^\top \mathbf{x}_t \in \mathbb{R}^{d_v}$ therefore requires a $KQ \times d_v$ weight matrix; at $Q = 5000$ and $K = 5$, this amounts to 1.6M parameters and a per-batch intermediate tensor of size batch $\times\, S \times 25000$, placing substantial memory pressure on GPU-constrained deployment environments.

We propose **Static Item Embedding** (SIE), an alternative value encoding that decouples item identity from response pattern via two independent additive branches:

$$\mathbf{v}_t = W_{\text{item}}\, \mathbf{u}_{q_t} + W_{\text{resp}}\, \mathbf{s}_t + \mathbf{b}_{\text{resp}} \in \mathbb{R}^{d_v}, \tag{4}$$

where $\mathbf{u}_{q_t} \in \mathbb{R}^H$ is a *frozen* (non-trainable) unit-norm random vector assigned to item q_t at initialization, $\mathbf{s}_t \in \mathbb{R}^K$ is the triangular ordinal code from (2), and $W_{\text{item}} \in \mathbb{R}^{d_v \times H}$ and $W_{\text{resp}} \in \mathbb{R}^{d_v \times K}$ are learned projections shared across all items.

The factored structure in (4) separates two gradient channels: W_{item} is updated by item-identity signals, while W_{resp} is updated by response-pattern signals. Because item vectors are frozen, items cannot collapse to similar representations during training, a failure mode of learned separable embeddings at large Q (as shown in the ablation study below). For two independent uniformly random unit vectors in $\mathbb{R}^H$, the expected squared inner product is $1/H$ (Dasgupta & Gupta, 2003), so pairwise interference concentrates around zero as H grows; cross-item gradient crosstalk in W_{item} therefore decays with H independently of Q, making SIE well-suited for large item banks.

The frozen dimension H is set as the smallest power of two no less than $\max(128, \lfloor Q/2 \rfloor)$, clamped to $[128, 1024]$.[1] This schedule ensures near-orthogonality scales with item bank size while keeping the projection dimension bounded, in contrast to the $\mathcal{O}(KQ)$ complexity of the full Kronecker encoding.

3.3 Dynamic Key-Value Memory Network

We adopt the Dynamic Key-Value Memory Network (DKVMN) as the sequential backbone. Let $\mathbf{k}_t = A^\top q_t \in \mathbb{R}^{d_k}$ be a key embedding for the current item and $\mathbf{v}_t \in \mathbb{R}^{d_v}$ be a value embedding for the interaction, obtained via the Kronecker projection $\mathbf{v}_t = B^\top \mathbf{x}_t$

[1] Concretely, $H = 128$ for $Q \leq 256$, $H = 256$ for $Q \approx 500$, $H = 512$ for $Q \approx 1000$, and $H = 1024$ for $Q \geq 2000$.

(ordinal-aware encoding) or the Static Item Embedding in (4) (scalable encoding). The model maintains a static key memory $\mathbf{M}^k \in \mathbb{R}^{N \times d_k}$ and a dynamic value memory $\mathbf{M}_t^v \in \mathbb{R}^{N \times d_v}$. Educationally, the key memory represents latent skill areas or knowledge components, the value memory tracks the student's evolving mastery of each area, and the attention weights $\mathbf{w}_t$ (below) indicate which skill areas a given item exercises. The summary vector $\mathbf{f}_t$ aggregates the student's current mastery state relevant to the item at hand.

Specifically, the key query is first projected via a learned transform, and attention weights over memory slots are computed as

$$\tilde{\mathbf{k}}_t = \tanh(W_{\text{attn}}\mathbf{k}_t), \qquad \mathbf{w}_t = \text{softmax}(\mathbf{M}^k\tilde{\mathbf{k}}_t) \in \mathbb{R}^N, \qquad w_t(i) = [\mathbf{w}_t]_i, \tag{5}$$

and the read vector is

$$\mathbf{r}_t = \sum_{i=1}^{N} w_t(i)\, \mathbf{M}_t^v(i) \in \mathbb{R}^{d_v}. \tag{6}$$

We fuse item context and retrieved state into a summary vector

$$\mathbf{f}_t = \tanh(W_{\mathbf{f}}[\mathbf{r}_t; \mathbf{k}_t] + b_{\mathbf{f}})\,, \tag{7}$$

which is then used by the response head to predict r_{t+1}.

Following memory-augmented neural networks, the value memory is updated via erase-add operations:

$$\mathbf{e}_t = \sigma(W_{\mathbf{e}}\mathbf{v}_t + b_{\mathbf{e}}), \tag{8}$$

$$\mathbf{a}_t = \tanh(W_{\mathbf{a}}\mathbf{v}_t + b_{\mathbf{a}}), \tag{9}$$

$$\mathbf{M}_{t+1}^v(i) = \mathbf{M}_t^v(i)\big[1 - w_t(i)\mathbf{e}_t\big] + w_t(i)\mathbf{a}_t. \tag{10}$$

3.4 DEEP-GPCM: an Ordinal IRT Head

Instead of the binary readout (e.g., $\sigma(W_p^\top \mathbf{f}_t + b_p)$), we use a Generalized Partial Credit Model (GPCM) head to obtain category probabilities for ordered responses. Let j denote the item exercised at time t (i.e., $[q_t]_j = 1$). We compute a time-varying ability proxy θ_t and item-specific parameters (thresholds and discrimination) as

$$\theta_t = W_\theta \mathbf{f}_t + b_\theta, \tag{11}$$

$$\beta_{j,0} = W_\beta \mathbf{k}_t + b_\beta, \tag{12}$$

$$\beta_{j,k} = \beta_{j,k-1} + \text{softplus}(W_{g[k]}\mathbf{k}_t + b_{g[k]}), \quad 1 \le k \le K-2, \tag{13}$$

$$\alpha_j = \exp(W_\alpha \mathbf{k}_t + b_\alpha)\,. \tag{14}$$

The construction in (13) enforces $\beta_{j,0} < \beta_{j,1} < \cdots < \beta_{j,K-2}$ by adding strictly positive increments (via softplus) to each preceding threshold. This monotonicity guarantee has direct educational meaning: achieving score level $k{+}1$ always requires higher ability than achieving level k, respecting the ordering that an assessment designer intended. Without this constraint, the model could learn disordered thresholds where some score

categories are never the most probable response at any ability level, a well-documented problem in educational measurement that signals a poorly functioning item (Wilson, 2005).

Because discrimination is an item-intrinsic property in the GPCM, α_j is derived solely from the item key $\mathbf{k}_t$, deliberately excluding the student-state summary $\mathbf{f}_t$. This separation ensures that the estimated discrimination reflects the item's inherent capacity to differentiate ability levels, independently of which student is currently being modelled — consistent with the IRT assumption that item parameters are population-invariant (Hambleton & Swaminathan, 1985). Item-level discrimination is recovered by averaging per-timestep estimates across all encounters with a given item (Meulders & Xie, 2004). The exponential activation ensures $\alpha_j > 0$.

Given $(\theta_t, \alpha_j, \{\beta_{j,k}\}_{k=0}^{K-2})$, the GPCM category probability is

$$p_{t,k} = \frac{\exp\Big(\sum_{i=0}^{k-1} \alpha_j(\theta_t - \beta_{j,i})\Big)}{\sum_{c=0}^{K-1} \exp\Big(\sum_{i=0}^{c-1} \alpha_j(\theta_t - \beta_{j,i})\Big)}. \tag{15}$$

At each timestep, the model evaluates this standard GPCM formulation using the *dynamically estimated* student ability θ_t and item parameters $\alpha_j, \boldsymbol{\beta}_j$. Unlike classical GPCM estimation where ability is static, here θ_t evolves as new interactions are observed, enabling the model to track learning over time while retaining the interpretable measurement structure of polytomous IRT.

We predict $\hat{r}_{t+1} = \arg\max_k p_{t,k}$, and optimize the weighted ordinal loss in (16) over the observed responses.

3.5 Training Objective for Ordinal Data

Because the GPCM defines category probabilities via (15) rather than cumulative links, we optimize a weighted ordinal loss (WOL) that jointly addresses class imbalance and ordinal distance (Gutiérrez, Pérez-Ortiz, Sánchez-Monedero, Fernández-Navarro, & Hervás-Martínez, 2016):

$$\mathcal{L}_{\text{WOL}} = \frac{1}{n} \sum_{i=1}^{n} \Big[(1 + \gamma |y_i - \hat{y}_i|) \cdot w_{y_i} \cdot (-\log \hat{p}_{i,y_i}) \Big], \tag{16}$$

where y_i is the true category, $\hat{p}_{i,y_i}$ is the predicted probability under (15), $\hat{y}_i = \arg\max_k \hat{p}_{i,k}$ is the predicted label, and γ controls the ordinal distance penalty. To balance classes, with c_k the sample count of class k, we set $w_k = \sqrt{\frac{n}{K \cdot c_k}}$, where n is the number of valid tokens in the batch. The distance term $(1 + \gamma|y_i - \hat{y}_i|)$ penalizes predictions that are far from the true category on the ordinal scale, encouraging the model to prefer near-misses over large errors—a property that standard cross-entropy does not enforce.

4 Experiments

4.0.1 Datasets

Synthetic-Ordinal.

To the best of our knowledge, there are no standardized public benchmarks for ordinal knowledge tracing. We therefore generate a synthetic dataset using an IRT-GPCM data generating process following established psychometric principles. This design provides controlled polytomous supervision $r_t \in \{0, \ldots, K-1\}$ and enables parameter recovery analysis with known ground truth.

We simulate 5000 students interacting with questions drawn from a pool of $Q = 200$ questions in random order. Each interaction sequence has variable length in $[20, 80]$. Students and questions are configured with *static* GPCM parameters following assumed priors:

- *Student ability:* $\theta \sim \mathcal{N}(0, 1)$,
- *Item discrimination:* $\alpha \sim \text{LogNormal}(0, 0.3)$,
- *Item thresholds:* $K-1$ values drawn independently from $\mathcal{N}(0, 1)$ and sorted to obtain $\beta_0 < \cdots < \beta_{K-2}$.

We generate five datasets with $K \in \{2, 3, 4, 5, 6\}$ to test extensibility across category resolutions, where $K = 2$ serves as the binary compatibility baseline and $K \in \{3, 4, 5, 6\}$ provide the ordinal evaluation conditions. Each dataset is split 80/20 for training/testing; a validation split is held out from training for early stopping and hyperparameter tuning.

Synthetic-Imbalanced.

Real educational assessments rarely produce uniformly distributed scores. High-performing students dominate most operational datasets, creating skewed response distributions where the highest category is most frequent. To test robustness to this common condition, we generate three imbalanced variants of the $Q = 200$, $K = 4$ dataset by shifting the student ability distribution upward: mild imbalance ($\theta \sim \mathcal{N}(0.5, 1)$), severe imbalance ($\theta \sim \mathcal{N}(1.0, 1)$), and extreme imbalance ($\theta \sim \mathcal{N}(1.5, 1)$). This produces progressively skewed response distributions where the highest category accounts for approximately 35%, 50%, and 65% of responses respectively, while retaining the same item parameters and GPCM data-generating process. All other settings match Synthetic-Ordinal.

Binary Compatibility

After validating the PKT instantiations on Synthetic-Ordinal, we confirm compatibility with binary KT. Due to implementation differences in batch handling and sequence-to-question lookup, we select two datasets:

- **ASSIST2015**[2]: 19,840 students and 100 KCs/questions.

[2] https://sites.google.com/site/assistmentsdata/datasets/2015-assistments-skill-builder-data

- **Synthetic-5**[3]: (Piech et al., 2015) simulated 4,000 students' responses with probability $p(r) = c + \frac{1}{1+\exp(-(\theta-\beta))}$ where $c = 0.25$ models question guessing. Each student is administered 50 questions in sequential order.

4.0.2 Models and Baselines

We compare DEEP-GPCM against three baselines that progressively relax psychometric and ordinal structure, each isolating a different design question:

- **DKVMN + Softmax**: DKVMN backbone with a K-way softmax head trained with categorical cross-entropy. No ordinal structure, no IRT constraints. Tests whether psychometric parameterization provides gains over a purely neural polytomous baseline.
- **Static GPCM**: Standard GPCM fit from interaction data with student ability held fixed across items. No sequential dynamics. Represents the psychometric status quo and quantifies what sequential modeling adds beyond classical IRT estimation.
- **Dynamic GPCM**: GPCM with per-item α/β lookup tables and a gated recurrent ability update driven by prediction error at each step. No DKVMN memory. Isolates the contribution of sequential ability updating from DKVMN's distributed memory.
- **DEEP-GPCM (ours)**: DKVMN backbone with GPCM head and weighted ordinal loss (16). Uses SIE encoding by default; the choice of encoding strategy is ablated below.

4.0.3 Prediction Mapping

For $K = 2$, predicted probabilities are converted to labels with a threshold of 0.5 following Zhang et al. (2017). For $K > 2$, we use $\hat{y}_t = \arg\max_k \hat{p}_{t,k}$ to map categorical probabilities to prediction labels.

4.0.4 Training Setup

We use the weighted ordinal loss in (16) to handle class imbalance and ordinal agreement.

We set $d_k = 64$ and $d_v = 128$ for key/value embedding dimensions and configure $N = 50$ memory slots for both memory matrices.

All models are trained with batch size 64 using Adam with learning rate 10^{-3}, reduced on performance plateau as per Mukherjee, Khare, and Verma (2019) for late-epoch stability. We apply dropout 0.05 and early stopping if validation loss does not improve for 10 epochs. Models are implemented in PyTorch and trained on an RTX 4060 Laptop GPU.

4.0.5 Evaluation Metrics

We use the following metrics:

[3] https://github.com/chrispiech/DeepKnowledgeTracing/tree/master/data/synthetic

- **Categorical Accuracy (ACC)**: percentage of exact matches between predicted category and ground truth. While straightforward, ACC does not distinguish near-misses from far-misses and is therefore insufficient as a sole ordinal metric.
- **Quadratic Weighted Kappa (QWK)** (Cohen, 1968): our primary metric for ordinal tasks. QWK applies a quadratic penalty proportional to the squared distance between predicted and true categories, making it the standard agreement measure in automated essay scoring and graded assessment. A QWK of 0.0 indicates chance-level agreement; 1.0 indicates perfect agreement.
- **Kendall's Tau (τ)**: measures rank-order agreement between predicted and true response sequences, penalizing all inversions equally regardless of distance.
- **Area Under the ROC Curve (AUC)**: used for the binary case ($K = 2$) to maintain comparability with prior KT literature.

4.0.6 IRT Analysis Setup

For parameter recovery analysis on Synthetic-Ordinal, we use:

- **$\boldsymbol{\theta}$ Aggregation**: we take the mean ability estimate $\bar{\theta} = \frac{1}{T}\sum_{t=1}^{T}\theta_t$ across all timesteps for each student, averaging out sequence-position noise while exploiting the full interaction history.
- **$\boldsymbol{\alpha}, \boldsymbol{\beta}$ Aggregation**: because questions appear with variable frequency due to random sequencing, we aggregate item parameters based on occurrences and use the sample mean for recovery analysis.

To enable fair comparison between estimated and true parameters, we apply IRT linking transformations that align the scales while preserving correlations. For student ability θ and item difficulty β, we use z-score normalization: $\theta_{\text{norm}} = (\theta - \mu_\theta)/\sigma_\theta$, where μ and σ are computed from the respective parameter sets (estimated or true). For discrimination α, we apply log-transformation followed by standardization: $\alpha_{\text{norm}} = \exp\big((\log\alpha - \mu_{\log\alpha})/\sigma_{\log\alpha} \cdot 0.3\big)$, where the 0.3 scaling factor matches the log-normal prior used in data generation ($\alpha \sim \text{LogNormal}(0, 0.3)$). This normalization is standard practice in IRT parameter recovery studies, as it removes arbitrary scale and location differences while preserving the correlation structure that reflects true parameter recovery quality.

4.1 Results

4.1.1 Ordinal Prediction on Synthetic-Ordinal (RQ1)

Table 1 compares DEEP-GPCM against all baselines across category resolutions. DEEP-GPCM achieves the highest QWK at every K, with the advantage widening at higher K. This trend suggests that an explicit ordinal prior becomes more valuable as the number of response boundaries grows: the GPCM head constrains $K-1$ ordered thresholds, whereas a softmax head must learn K independent logits without structural guidance.

The baseline comparisons isolate the contribution of each design element. Static GPCM's low QWK confirms that psychometric structure alone is insufficient without sequential dynamics: a fixed ability cannot adapt to evolving learner states.

Table 1 Ordinal prediction on Synthetic-Ordinal ($Q = 200$). DEEP-GPCM outperforms all baselines on QWK. Best per column in bold.

	$K = 3$		$K = 4$		$K = 5$		$K = 6$	
Model	**ACC**	**QWK**	**ACC**	**QWK**	**ACC**	**QWK**	**ACC**	**QWK**
Static GPCM	.407	.168	.310	.156	.262	.178	.216	.150
Dynamic GPCM	.549	.533	.459	.619	.389	.681	.337	.704
DKVMN+Softmax	**.597**	.578	**.531**	.667	**.511**	.740	**.476**	.773
DEEP-GPCM	.596	**.592**	.529	**.682**	.507	**.759**	.470	**.791**

Dynamic GPCM closes much of this gap through recurrent updates, but still falls short of models with distributed memory. The comparison between DEEP-GPCM and DKVMN+Softmax is most revealing: both share the same DKVMN backbone, so their difference isolates the effect of the response head. The GPCM head's ordinal constraint acts as an inductive bias that regularizes learning, particularly at higher K where the softmax must partition a larger probability space without structural guidance. Notably, this structure improves prediction without limiting expressiveness—the GPCM head constrains the parameterization, not the set of distributions that can be expressed.

4.1.2 Binary Compatibility on Standard KT Benchmarks

A practical requirement for any polytomous KT framework is backward compatibility with existing binary benchmarks. When $K = 2$, the GPCM category probability in (15) reduces to a single logistic function of $\alpha_j(\theta_t - \beta_{j,0})$, recovering the standard 2PL IRT model. Table 2 confirms that this theoretical equivalence holds in practice: DEEP-GPCM matches DKT and DKVMN on ASSIST2015 and Synthetic-5, with AUC differences within 0.007. The psychometric parameterization imposes no measurable cost when ordinal structure is absent.

Table 2 Binary compatibility ($K = 2$). DEEP-GPCM matches neural baselines. Best per column in bold.

Model	**Synthetic-Ordinal**		**ASSIST2015**		**Synthetic-5**	
	ACC	**QWK**	**ACC**	**AUC**	**ACC**	**AUC**
DKT	—	—	75.2%	0.727	**75.2**%	0.817
DKVMN	—	—	75.1%	**0.729**	75.1%	**0.827**
DEEP-GPCM	**70.6**%	**0.410**	**75.4**%	0.723	74.7%	0.824

4.1.3 Learner State Dynamics (RQ2)

Figure 2 visualizes estimated ability trajectories θ_t for four learner archetypes: consistently high-performing, consistently struggling, improving, and declining. Two findings are central.

First, DEEP-GPCM and DKVMN+Softmax produce nearly identical trajectories across all archetypes, confirming that the GPCM head functions as a measurement lens: it re-expresses the backbone's learned dynamics in psychometric terms (θ_t, α_j, $\boldsymbol{\beta}_j$) without distorting the underlying sequential signal. This property is important for adoption because it means practitioners gain interpretable parameters without retraining or changing the backbone architecture.

Second, Dynamic GPCM's trajectories remain flat near $\theta \approx 0$ regardless of archetype. Without distributed memory to separate item characteristics from learner mastery, the recurrent update conflates the two signals: the model cannot distinguish whether a correct response reflects high ability or low item difficulty, so θ collapses to the population mean. This suggests that DKVMN's key–value separation is a structural prerequisite for psychometric identifiability in sequential settings, not merely a computational convenience.

4.1.4 IRT Parameter Recovery (RQ3)

Table 3 reports recovery correlations across all three parameter types and category resolutions. The results reveal a characteristic recovery profile for each model that reflects its architectural properties.

Student ability θ.

DEEP-GPCM achieves $r_\theta \geq 0.92$ across all K, approaching the reliability standards for operational adaptive testing (Reckase, 2009), with recovery improving slightly at higher K as finer ordinal resolution provides more information per interaction. Static GPCM recovers ability moderately ($r_\theta \approx 0.60$): fixed parameters capture population distributions but not individual trajectories. Dynamic GPCM's near-zero r_θ confirms the identifiability collapse observed in Section 4.1.3.

Item discrimination α.

DEEP-GPCM recovers discrimination at $r_\alpha > 0.70$ across all K despite never observing ground-truth α during training. This implicit recovery emerges because the GPCM parameterization (Eq. 14) forces item informativeness to be represented explicitly through α_j, while the memory network provides sufficient capacity to learn item-specific discrimination from response variation.

Step thresholds β.

A clear tradeoff appears: Static and Dynamic GPCM achieve near-perfect threshold recovery ($\bar{r}_\beta > 0.96$) through direct lookup tables with unmediated gradient paths. DEEP-GPCM trades some threshold precision ($\bar{r}_\beta \approx 0.91$) for the ability to model sequential dynamics, since thresholds must be inferred through memory reads. This

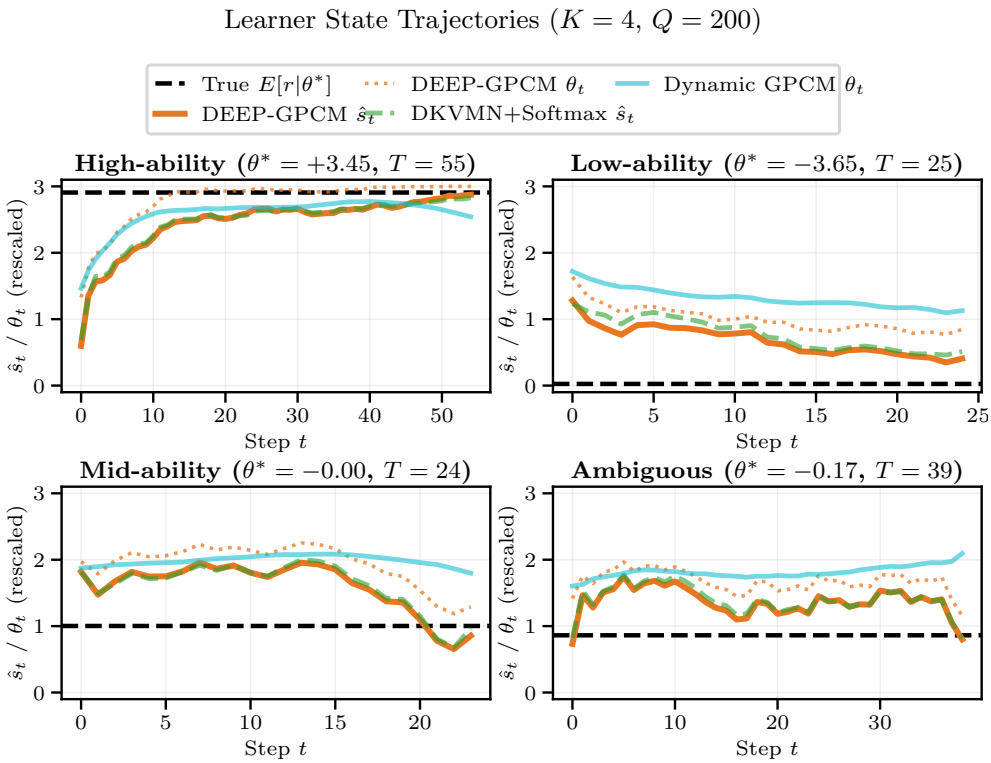

Fig. 2 Learner ability trajectories for four archetypes at $K = 4$, $Q = 200$. DEEP-GPCM differentiates high/low ability and growth/decline patterns. DKVMN+Softmax produces identical trajectories, confirming the GPCM head preserves dynamics. Dynamic GPCM fails to differentiate archetypes.

level remains above the $r > 0.85$ benchmark accepted in operational IRT calibration (Embretson & Reise, 2000).

Figures 3 and 4 visualize recovery at $K = 4$, $Q = 200$. DEEP-GPCM's estimates cluster tightly around the identity line for both student and item parameters, and the KDE plots show that estimated ability distributions closely match the ground-truth $\mathcal{N}(0, 1)$ prior.

DEEP-GPCM is the only model that achieves strong recovery across all three parameter types simultaneously. Static GPCM excels at item parameters but cannot track dynamic ability; Dynamic GPCM captures neither reliably. This balanced recovery enables DEEP-GPCM to serve a dual role that neither baseline can fulfill: predicting future responses while producing interpretable psychometric parameters for item analysis and instructional decision-making.

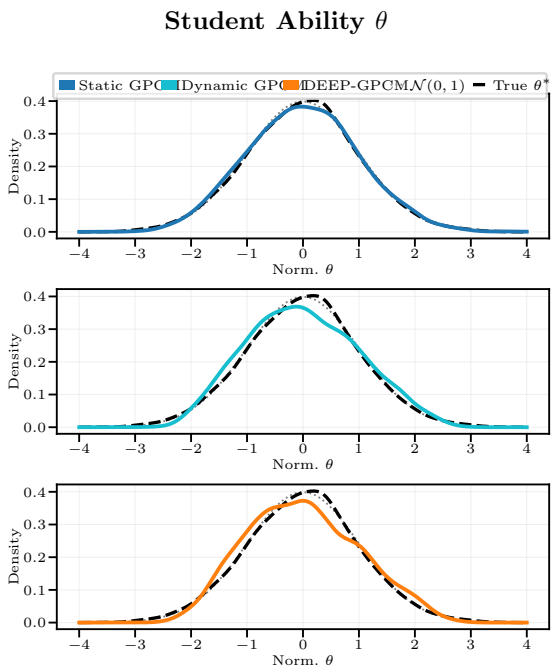

Fig. 3 Student ability θ recovery at $K = 4$, $Q = 200$. DEEP-GPCM achieves $r_\theta = 0.94$, approaching operational testing standards. Static GPCM ($r_\theta = 0.58$) captures population distributions but not individual trajectories. Dynamic GPCM ($r_\theta = 0.06$) cannot disentangle ability from item characteristics.

Table 3 Parameter recovery across models and K at $Q = 200$. DEEP-GPCM recovers discrimination without supervision. Best per column in bold.

	$K = 3$			$K = 4$			$K = 5$			$K = 6$		
Model	r_α	$\bar{r}_\beta$	r_θ	r_α	$\bar{r}_\beta$	r_θ	r_α	$\bar{r}_\beta$	r_θ	r_α	$\bar{r}_\beta$	r_θ
Static GPCM	.700	**.989**	.60	.606	**.983**	.58	.698	**.985**	.62	.539	**.984**	.61
Dynamic GPCM	.645	.976	.05	.700	.966	.06	.694	.965	.06	.523	.960	.07
DEEP-GPCM	**.754**	.816	**.92**	**.707**	.918	**.94**	**.746**	.912	**.95**	**.759**	.927	**.95**

4.1.5 Ablation and Sensitivity Analysis

Monotonic threshold constraint.

Table 4 shows that enforcing $\beta_0 < \beta_1 < \cdots < \beta_{K-2}$ through softplus gaps (Eq. 13) substantially improves threshold recovery with negligible effect on prediction. Without this constraint, the model can learn "disordered thresholds"—orderings where some score categories are never the most probable response at any ability level—which undermines interpretability (Wilson, 2005). Monotonicity thus acts as a theoretically

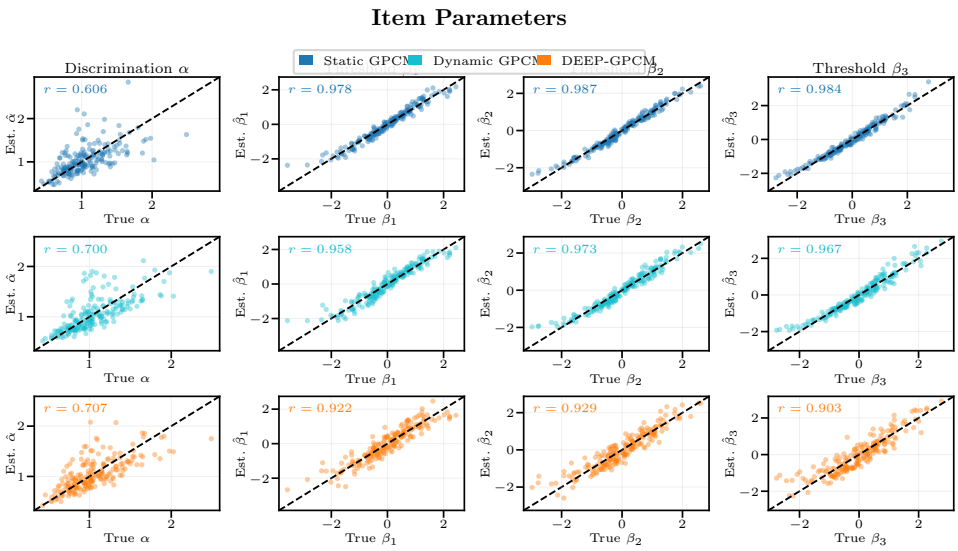

Fig. 4 Item parameter recovery (α, β) at $K = 4$, $Q = 200$. DEEP-GPCM achieves $r_\alpha = 0.707$ and $\bar{r}_\beta = 0.918$ through implicit psychometric inference from sequential interaction patterns.

motivated regularizer, eliminating psychometrically invalid parameter configurations without limiting the model's capacity.

Table 4 Monotonicity constraint ablation at $Q = 200$, $K = 4$. Constraint improves threshold recovery by 34 points. Best per column in bold.

Model	QWK	τ	r_α	$\bar{r}_\beta$	r_θ
Monotonic	**.682**	**.513**	**.707**	**.918**	**.94**
Unconstrained	.679	.510	.676	.684	.93

Item encoding scalability.

Table 5 reveals an interaction between encoding strategy and item bank size. At small scales ($Q \leq 500$), Separable encoding achieves the strongest discrimination recovery, with SIE close behind. As item banks grow, SIE overtakes Separable and reaches the highest r_α at $Q = 2000$. This crossover reflects how each encoding handles inter-item interference: learned embeddings (Separable) are susceptible to gradient crosstalk at large Q, where updates for one item inadvertently shift neighbouring representations. SIE's frozen random projections are near-orthogonal by construction, so item representations remain distinct regardless of bank size. LinDecay's recovery is inconsistent across scales, suggesting sensitivity to interactions between the high-dimensional

Kronecker representation, item diversity, and sample coverage—factors whose joint influence warrants further investigation. Prediction quality (QWK) and student ability recovery (r_θ) are stable across encodings and scales, so encoding choice primarily affects item parameter recovery.

Table 5 Item encoding scalability at $K = 4$. Best per column in bold within each Q.

$Q = 200$					$Q = 500$				
Enc.	**QWK**	r_α	$\bar{r}_\beta$	r_θ	**Enc.**	**QWK**	r_α	$\bar{r}_\beta$	r_θ
LinDecay	**.684**	.319	.861	**.95**	LinDecay	.697	.419	.818	**.96**
Separable	**.684**	**.739**	.872	**.95**	Separable	**.701**	**.799**	.822	**.95**
SIE	.682	.707	**.918**	.94	SIE	.699	.785	**.902**	.94
$Q = 1000$					$Q = 2000$				
Enc.	**QWK**	r_α	$\bar{r}_\beta$	r_θ	**Enc.**	**QWK**	r_α	$\bar{r}_\beta$	r_θ
LinDecay	.691	.198	.830	**.95**	LinDecay	.708	.802	.834	**.98**
Separable	**.696**	**.718**	**.913**	**.95**	Separable	.711	.773	.873	**.98**
SIE	.693	.618	.876	.93	SIE	**.712**	**.856**	**.924**	**.98**

Class imbalance robustness.

Operational assessments typically exhibit skewed response distributions where higher score categories predominate. Table 6 evaluates DEEP-GPCM under progressively severe imbalance at $Q = 200$, $K = 4$. The three parameter types degrade at different rates. **Student ability** is most robust, remaining above the 0.80 threshold for individual-level measurement (Embretson & Reise, 2000) even under extreme imbalance, because the DKVMN accumulates evidence across the full interaction sequence. **Step thresholds** degrade moderately; the monotonicity constraint (Eq. 13) anchors threshold ordering even when some categories have few observations. **Item discrimination** is most sensitive: estimating how sharply an item discriminates requires response variation across ability levels, and severe skew undersamples the lower ability range. This differential sensitivity means that student ability tracking tolerates substantial skew, but item calibration benefits from balanced response distributions.

5 Discussion

5.1 Educational Implications

DEEP-MIRT extends binary KT to address wider applications for formative assessment practice. By estimating a dynamic ability and categorical probabilities, it differentiates partial knowledge from total misunderstanding, providing richer insights in continuous learning progressions. The model's ability recovery ($r_\theta = 0.94$) meets operational standards for adaptive testing: traditional IRT calibration studies require

Table 6 Class imbalance robustness at $Q = 200$, $K = 4$. Response distribution shows empirical proportions for categories 0–3.

Condition	Distribution	QWK	r_α	r_β	r_θ
Balanced	.31/.21/.20/.29	0.683	0.777	0.927	0.941
Mild	.16/.16/.21/.46	0.669	0.736	0.909	0.931
Severe	.08/.11/.20/.62	0.630	0.641	0.844	0.896
Extreme	.05/.08/.18/.69	0.594	0.543	0.829	0.877

20–40 items per student to achieve comparable reliability, but DEEP-GPCM achieves this dynamically from sequential interaction data. This enables real-time ability tracking without requiring students to complete lengthy calibration assessments.

Furthermore, the extracted item parameters describe item-level characteristics beyond immediate predictive use. Unlike classical psychometric calibration, DEEP-MIRT allows online extraction of item characteristics alongside student states, which helps identify underperforming or misaligned items in real time. For example, items with unexpectedly low discrimination ($\alpha < 0.5$) may signal ambiguous wording or construct-irrelevant difficulty, while disordered thresholds indicate score categories that need revision (Wilson, 2005). This transition from a purely predictive instrument to a measurement-aware system ensures that instructional decisions are grounded in identifiable constructs such as discrimination and step difficulty (Heritage, 2007).

5.2 Limitations and Future Directions

While the framework establishes a foundation for polytomous knowledge tracing, several limitations present direct avenues for future research.

First, although we proposed DEEP-MIRT as a modular framework with interchangeable components, the current implementation DEEP-GPCM is limited to the GPCM head. A significant research opportunity lies in fully realizing this modularity. For example, a nominal response head using the NRM (Bock, 1972) would extend the framework to unordered polytomous responses such as multiple-choice distractors, enabling misconception-level feedback.

Second, evaluating parameter recovery on synthetic data generated under GPCM assumptions is inherently circular, and proxy signals like attempt counts contain construct-irrelevant variance. Future evaluations will apply the model to authentic polytomous datasets and validate the inferred trajectories against external standardized measures.

Third, the current backbone maintains a unidimensional student state and does not address multi-dimensional discrimination recovery. Connecting multidimensional instantiations with concept-aligned memory slots would enable per-skill ability tracking, supporting more targeted diagnostics than a composite θ_t.

6 Conclusion

The transition from binary to polytomous responses is essential for practical educational assessment. DEEP-MIRT reconciles the predictive performance of sequential

neural networks with the rigorness of IRT. By decoupling the sequential backbone from interchangeable psychometric heads, we demonstrate that deep learning can optimize prediction while yielding identifiable measurement constructs. The framework's backward compatibility and scalability confirm its viability as a practical replacement for existing systems. By respecting the ordinal structure of educational data, DEEP-MIRT establishes an alternative for measurement-aware adaptive learning.

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Appendix A Additional Parameter Recovery Plots

Figures A1–A6 present parameter recovery plots for $K \in \{3, 5, 6\}$ at $Q = 200$, supplementing the $K = 4$ results shown in Figures 3 and 4. Each row corresponds to one model (Static GPCM, Dynamic GPCM, DEEP-GPCM from top to bottom).

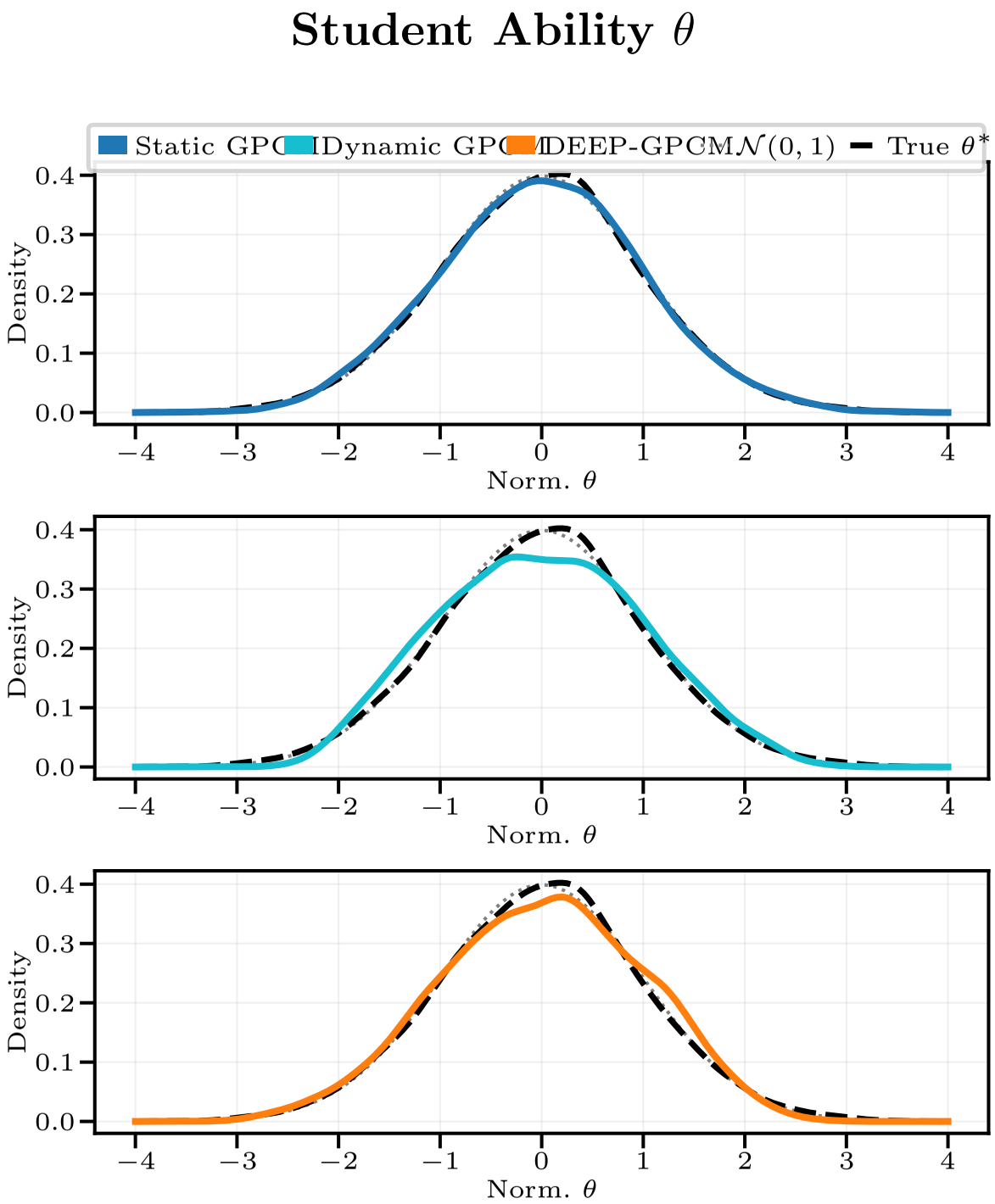

Fig. A1 Student ability (θ) recovery at $K = 3$, $Q = 200$.

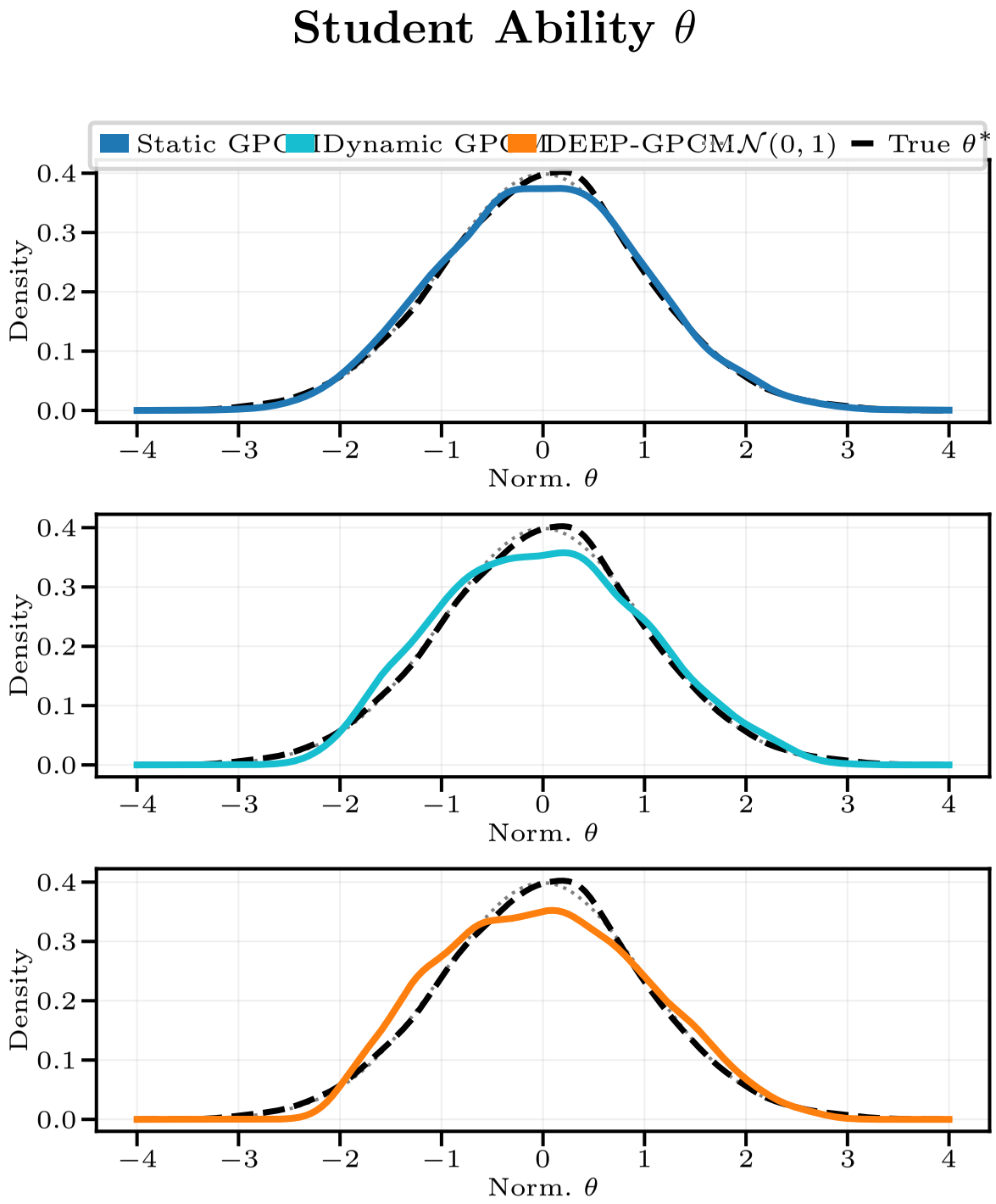

Fig. A2 Student ability (θ) recovery at $K = 5$, $Q = 200$.

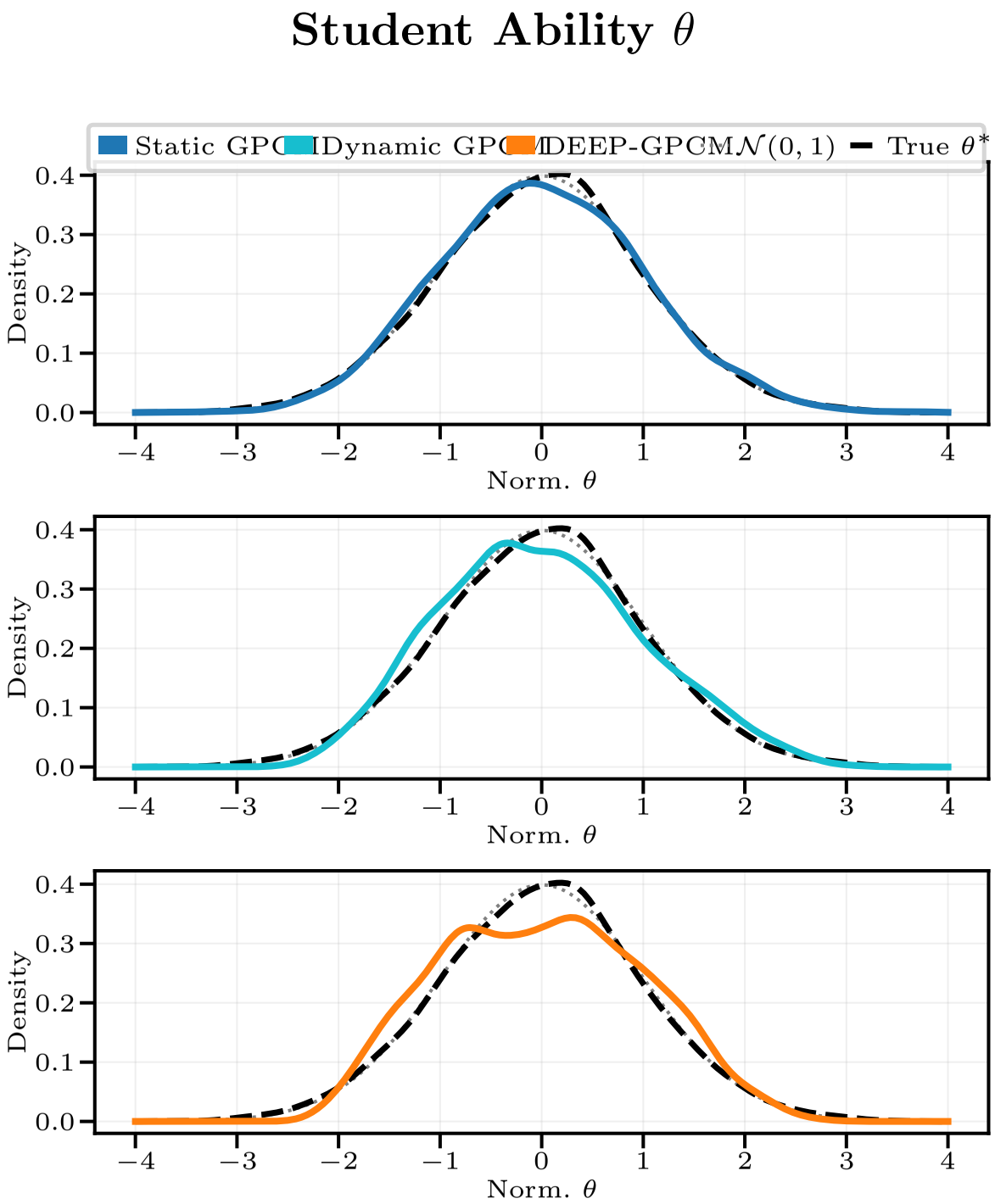

Fig. A3 Student ability (θ) recovery at $K = 6$, $Q = 200$.

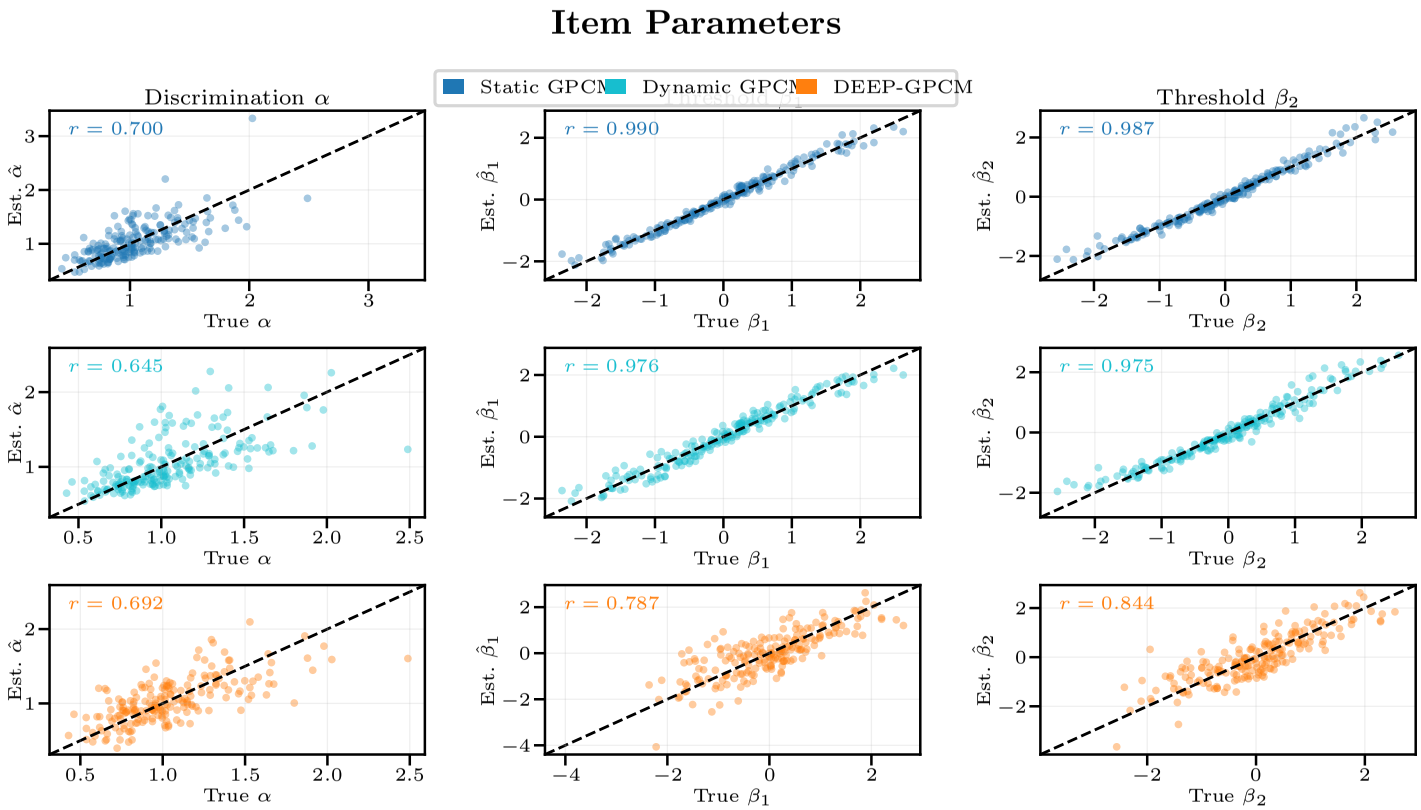

Fig. A4 Item parameter (α, β) recovery at $K = 3$, $Q = 200$.

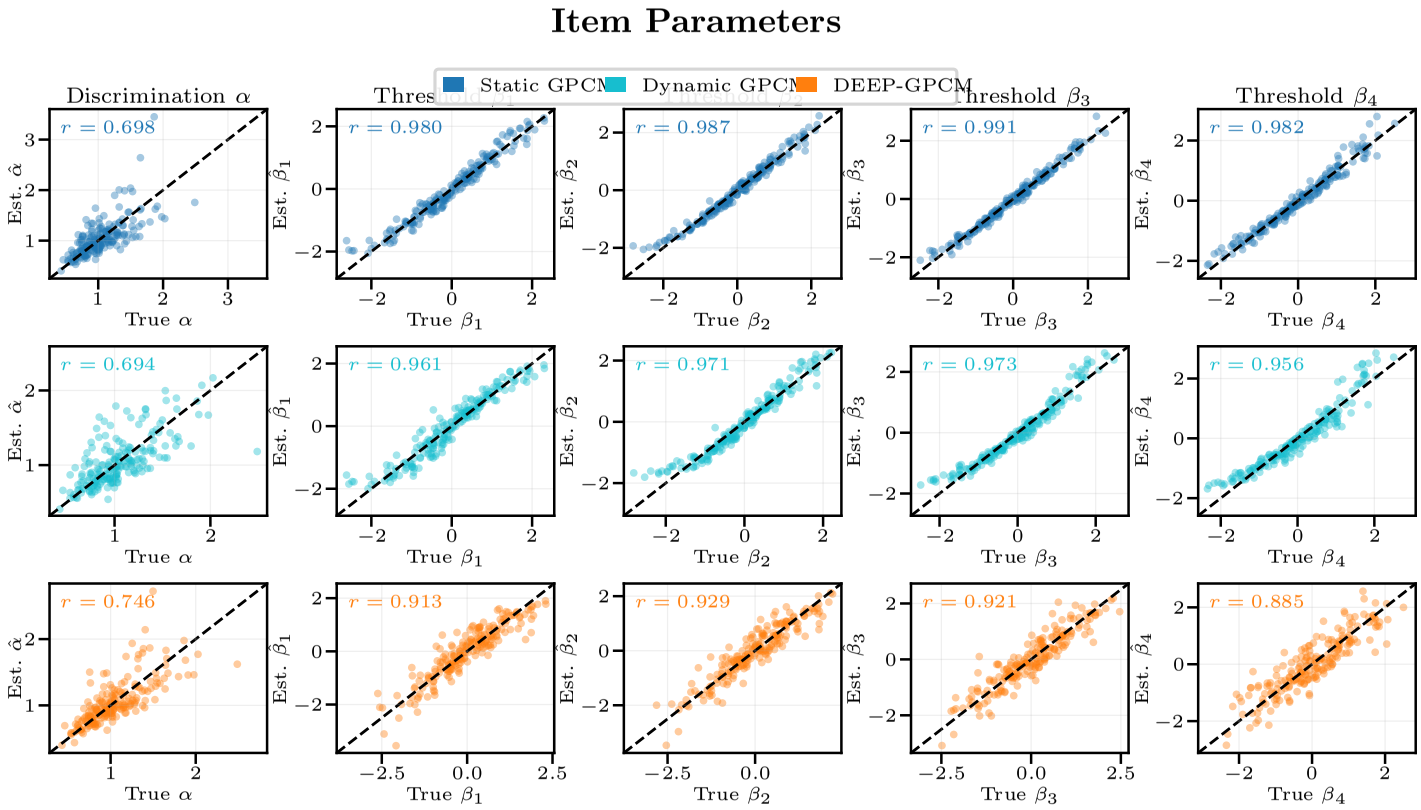

Fig. A5 Item parameter (α, β) recovery at $K = 5$, $Q = 200$.

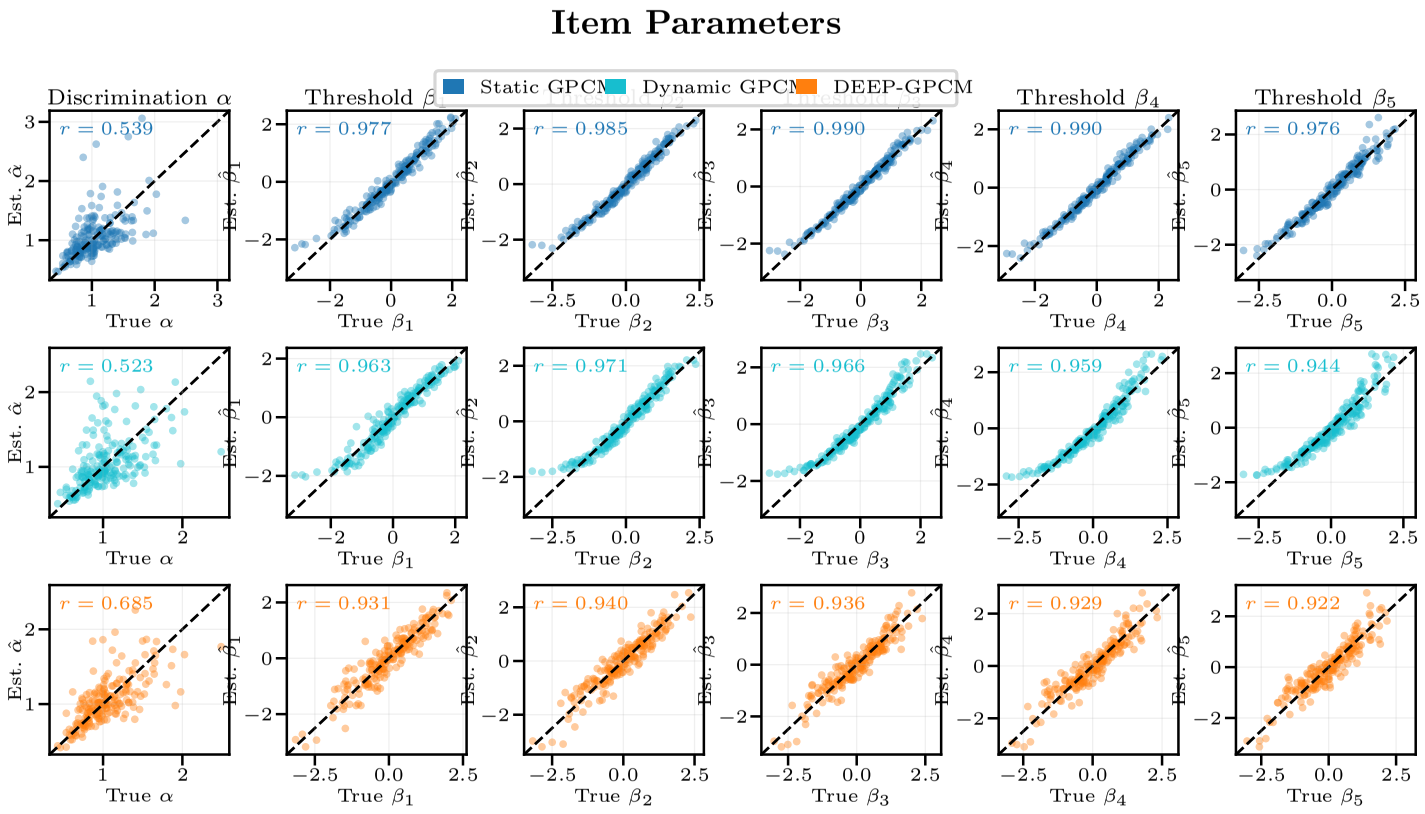

Fig. A6 Item parameter (α, β) recovery at $K = 6$, $Q = 200$.